INSPIRE: An Internalize-Then-Improve Approach for Example-Driven Mathematical Reasoning

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Abstract

Mathematical reasoning has seen rapid progress in large language models (LLMs), yet existing methods optimize predominantly for final-answer correctness, raising the question whether models truly internalize mathematical concepts or merely memorize solution patterns. In human mathematics education, example-based reasoning such as constructing counterexamples to test theorem boundaries reflects deep conceptual understanding, but remains underdeveloped in current LLMs. Enhancing this capability through preference optimization presents two key challenges: (1) the model's limited example-based reasoning ability makes constructing effective preference pairs inherently difficult; and (2) capability acquisition is progressive, as the model must first learn to adopt this strategy before learning to apply it correctly. Therefore we propose INSPIRE, an Internalize-Then-Improve approach combining Reference-Guided Student Internalization (RGSI), which produces high-quality preference candidates under the policy model's own distribution, with a stage-wise rubric preference training strategy that decomposes learning into method-oriented and correctness-oriented stages. Experiments across multiple model scales and families demonstrate consistent improvements, even surpassing larger open-source models, while evaluations on out-of-distribution benchmarks confirm no degradation in general mathematical reasoning ability.

1 Introduction

Mathematical reasoning is a core aspect of intelligence and a major focus of large language model (LLM) research. (Singh et al., 2025; Guo et al., 2025; Yang et al., 2024; Lu et al., 2025). Enhancing the mathematical reasoning capability of LLMs has become a prominent and fundamental research topic within the community (Xu et al., 2026; Zhang et al., 2026; An et al., 2026; Li et al., 2026a).

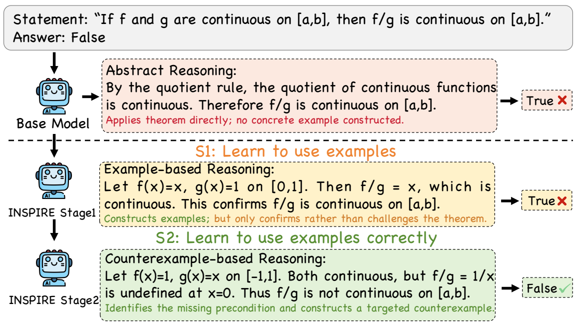

Figure 1: Overview of INSPIRE's progressive training effect on a mathematical statement requiring example-driven reasoning.

Substantial recent efforts have been devoted to this goal, including data augmentation (Toshniwal et al., 2024; Ye et al., 2025), supervised fine-tuning (Ying et al., 2024; Li et al., 2025a; Muennighoff et al., 2025), and reinforcement learning-based post-training (Shao et al., 2024; Du et al., 2025). Despite their success, most methods optimize for final-answer correctness, raising the question of whether models truly internalize mathematical concepts or rely on surface-level solution patterns.

In human mathematics education, example-based reasoning is a fundamental skill reflecting deep conceptual understanding (Klymchuk, 2010). Recent work has shown that this capability remains underdeveloped in current LLMs (Wu et al., 2024; Li et al., 2025b; Kuang et al., 2025; Li et al., 2026b). As illustrated in Figure 1, models often rely on abstract theorem application rather than constructing concrete examples to test their reasoning, a tendency reinforced by training paradigms that optimize predominantly for final-answer correctness.

Enhancing this capability through preference optimization presents two key challenges: **Data construction under ability scarcity.** Since the model exhibits limited example-based reasoning

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ability, self-sampled candidates are generally low in quality on this dimension, failing to form effective preference pairs. Directly using external references as preferred samples introduces distribution shift in both expression style and reasoning structure, causing the model to replicate surface-level characteristics rather than internalize core analytical strategies. **Progressive nature of capability acquisition.** Since the model exhibits limited example-based reasoning ability, it must first learn to actively adopt this strategy, and only then to apply it correctly, making it difficult to optimize for both aspects in a single step.

To address these challenges, we propose INSPIRE, an **IN**ternalize-then-improve approach using **S**tage-wise **P**reference for **I**mproved mathematical **RE**asoning. The internalization stage introduces Reference-Guided Student Internalization (RGSI), which enables the model to comprehend reference reasoning approaches and regenerate responses in its own expression style, producing high-quality preference candidates while avoiding distribution shift. Built upon this, we design a stage-wise preference training strategy that first encourages the model to actively adopt example-based reasoning, and then progressively refines answer correctness on top of the acquired strategy. Our contributions are summarized as follows:

- We propose INSPIRE, an Internalize-Then-Improve approach that enhances example-driven mathematical reasoning through preference optimization, addressing both the absence of this ability and its progressive nature of acquisition.
- INSPIRE introduces Reference-Guided Student Internalization to produce high-quality preference candidates, and a stage-wise rubric preference training strategy that decomposes learning into method-oriented and correctness-oriented.
- Experiments across multiple model scales and families demonstrate consistent improvements, with comprehensive ablations and analyses validating each design choice. Our code has been released at `https://github.com/Shea-code-xxx/INSPIRE-code`.

2 Related Work

Mathematical Reasoning in LLMs. Existing approaches generally follow two lines of work. The first leverages supervised fine-tuning and data augmentation to strengthen problem-solving ability, scaling up training data through instruction synthesis, question rephrasing, and corpus expansion (Luo et al., 2025; Yu et al., 2024; Li et al., 2024a). The second employs reinforcement learning for post-training to elicit deeper reasoning capabilities. Representative efforts include GRPO (Shao et al., 2024), OpenAI o1 (Jaech et al., 2024), DeepSeek-R1 (Guo et al., 2025), and DAPO (Yu et al., 2025), which demonstrate that large-scale reinforcement learning can induce self-verification and reflection behaviors in models. Despite their success, these methods uniformly optimize for final-answer correctness, paying limited attention to deeper reasoning skills such as concept understanding and discrimination through examples and counterexamples (Klymchuk, 2010). Recent benchmarks such as ConceptMath (Wu et al., 2024) and CounterMath (Li et al., 2025b) reveal considerable room for improvement in concept-level mathematical reasoning, yet how to effectively enhance this ability through targeted training that guides models to actively adopt analytical strategies such as example-based reasoning remains largely unexplored.

Preference Optimization for Reasoning. Direct Preference Optimization (DPO; Rafailov et al., 2023) and its variants such as IPO (Azar et al., 2024) and SimPO (Meng et al., 2024) have been applied to mathematical reasoning. Step-DPO (Lai et al., 2024) and Full-Step-DPO (Xu et al., 2025) treat individual reasoning steps as the basic unit for preference optimization, achieving notable results on long-chain reasoning tasks. Math-Shepherd (Wang et al., 2024) trains a process reward model to provide step-level supervision, moving beyond binary outcome-level signals. More recently, rubric-based approaches (Gunjal et al., 2025; Huang et al., 2025) have shown that decomposing evaluation into multiple structured criteria can provide richer training signals than single-dimensional rewards. A key challenge is constructing high-quality preference data under the policy model's own distribution. Common strategies include rejection sampling (Singh et al., 2024; Yuan et al., 2024) and iterative online DPO that refreshes training pairs with updated model outputs (Xu et al., 2024). However, when the model has limited target reasoning ability, on-policy sampling alone cannot produce sufficient quality variation, while off-policy data from stronger models risks distribution shift. Moreover, these methods generally rely on answer correctness as the sole preference signal, leaving multi-dimensional evaluation and progressive training strategies largely unexplored.

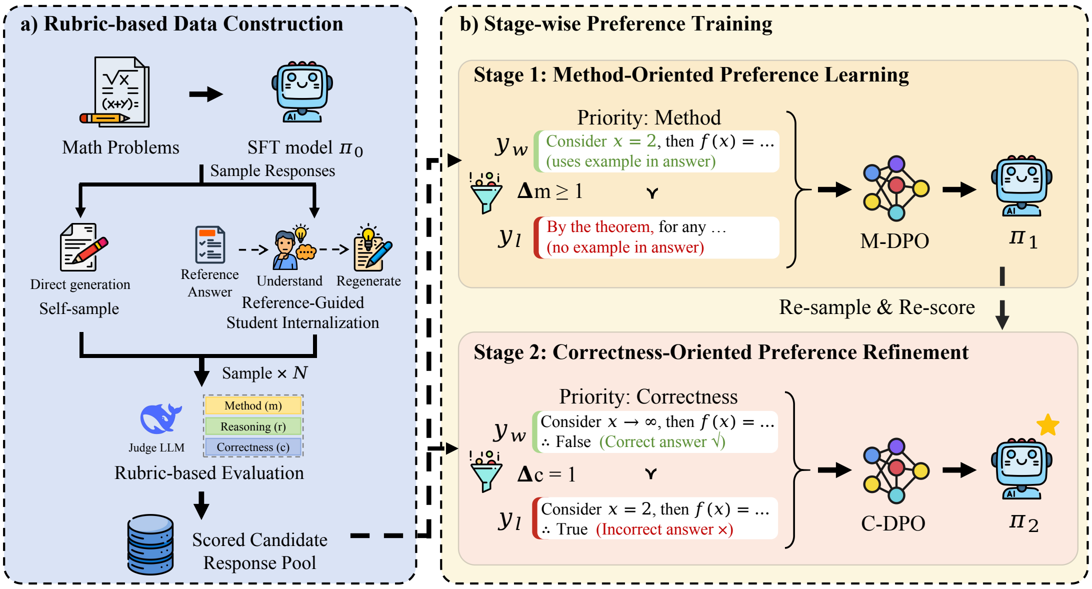

Figure 2: Overview of INSPIRE. (a) **Rubric-based Data Construction**: candidate responses are obtained via both self-sampling and Reference-Guided Student Internalization (RGSI) from the base model, then scored by a judge LLM along three dimensions (Method, Reasoning, Correctness). (b) **Stage-wise Preference Training**: Stage 1 constructs preference pairs prioritizing the method dimension to encourage example-based reasoning; Stage 2 re-samples from the Stage 1 model and constructs pairs prioritizing answer correctness, with Stage 1 replay samples mixed in to prevent forgetting. Note that both responses in Stage 2 employ examples, reflecting the method acquisition from Stage 1; the preference signal now targets correctness.

Dimension	**Range**	**Description**
Method (m)	0–2	Use of example-based analysis
Reasoning (r)	0–3	Logical quality of reasoning
Correctness (c)	0–1	Correctness of the final answer

Table 1: Three-dimensional evaluation rubric.

3 Methodology

In this section, we present INSPIRE in detail. Built around a multi-dimensional evaluation rubric, our method first constructs high-quality preference data under the policy model's own distribution (§3.1), and then applies a stage-wise training strategy that progressively acquires the target reasoning capability (§3.2). Figure 2 illustrates the overall approach.

3.1 Rubric-based Preference Data Construction

Constructing high-quality preference data requires addressing two issues: how to evaluate response quality to distinguish preferred from dispreferred samples, and how to obtain sufficient training pairs with meaningful quality differences. We describe our multi-dimensional evaluation rubrics and data construction strategy in turn.

Evaluation Rubric. Most existing preference optimization methods for mathematical reasoning adopt answer correctness as the sole preference signal. However, a single correctness signal is insufficient for our setting, as it cannot distinguish whether the model actively adopts example-based reasoning or merely arrives at the answer through abstract theorem application. Inspired by recent rubric-based training approaches (Gunjal et al., 2025; Huang et al., 2025) that demonstrate the effectiveness of decomposing evaluation into multiple structured criteria, and drawing on standard practices in mathematics education where student proofs are assessed from multiple perspectives, we design a three-dimensional rubric: **Method** (whether the response adopts an example-based analytical strategy), **Reasoning** (the logical coherence and completeness of the argument), and **Correctness** (whether the final answer is correct). We design dimension-specific scoring criteria, as summarized in Table 1 (detailed in Appendix G). Following the LLM-as-a-Judge paradigm (Zheng et al., 2023; Li et al., 2024b, 2025c; Kuang et al., 2026; Liu et al., 2026), we employ DeepseekV3.2 (Liu et al., 2024) to score each response independently along the three dimensions. The full prompts are provided in Appendix A.

Response Augmentation via Reference-Guided

Student Internalization. Effective preference learning requires candidate responses with meaningful quality differences. Meanwhile, preference optimization methods such as DPO benefit from training data generated under the policy model's own distribution (Rafailov et al., 2023; Li et al., 2022; Xu et al., 2024; Shen et al., 2026), as off-policy data introduces distribution mismatch that degrades training effectiveness. These two requirements create a tension in our setting: since the base model exhibits limited example-based reasoning ability, on-policy sampling alone produces candidates that are uniformly low in quality on this dimension, failing to provide sufficient quality gradients for preference pair construction. To resolve this tension, we propose **Reference-Guided Student Internalization (RGSI)**. In contrast to standard self-sampling where candidates are drawn from $\pi_0(\cdot \mid x)$, RGSI conditions the generation on both the problem x and the reference solution r:

$$\tilde{y} \sim \pi_0(\cdot \mid x, r), \tag{1}$$

where r provides reasoning guidance while the policy model π_0 serves as the generator, ensuring that the output remains on-policy in distribution. Meanwhile, informed by the reference, the generated responses exhibit higher quality in method usage and reasoning than purely self-sampled candidates, providing the necessary quality gradients. Implementation details are provided in Appendix C. We validate the effectiveness of this strategy through ablation studies in Section 4.4.

3.2 Stage-wise Rubric Preference Training

We build our training strategy upon Direct Preference Optimization (DPO; Rafailov et al., 2023). Since the base model lacks example-based reasoning ability, jointly optimizing for method acquisition and answer correctness in a single stage is suboptimal (verified empirically in Section 4.4). We therefore decompose training into two stages: **M-DPO** (Method-oriented DPO), which encourages the model to actively adopt example-based reasoning, and **C-DPO** (Correctness-oriented DPO), which refines answer correctness based on the acquired example-based reasoning strategy.

Stage 1:Method-Oriented Preference Learning The first stage aims to encourage the model to actively adopt example-based reasoning. We construct preference pairs primarily based on the method dimension. Specifically, for the set of candidate responses to each problem, a preference pair (y_w, y_l) must satisfy:

$$\begin{cases} m(y_w) - m(y_l) \geq 1 \\ r(y_w) \geq 1,\ r(y_l) \geq 1 \end{cases}, \tag{2}$$

where $m(\cdot)$ and $r(\cdot)$ denote the method and reasoning scores, respectively. The first constraint ensures a clear separation between responses that employ example-based reasoning and those that do not. The latter two serve as quality gates, filtering out responses with incoherent reasoning. Among all valid pairs, we further prioritize those with higher $r(y_w)$, exposing the model to high-quality reasoning exemplars while acquiring the target strategy. The M-DPO training objective is:

$$\begin{aligned} \mathcal{L}_{\text{M-DPO}} = -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}_1}\Bigg[\log\sigma\bigg(\beta_m \log\frac{\pi_\theta(y_w \mid x)}{\pi_0(y_w \mid x)} \\ - \beta_m \log\frac{\pi_\theta(y_l \mid x)}{\pi_0(y_l \mid x)}\bigg)\Bigg], \end{aligned} \tag{3}$$

where π_0 is the SFT model serving as both the initialization and the reference policy, β_m is the temperature parameter, and $\mathcal{D}_1$ denotes the set of method-oriented preference pairs. The resulting model π_1 is used as the starting point for Stage 2.

Stage 2:Correctness-Oriented Preference Refinement. After Stage 1, the model has acquired a preliminary ability to employ example-based reasoning. The second stage shifts the optimization focus toward answer correctness. We first use π_1 to re-sample candidate responses for each problem and score them using the same evaluation rubric.

The core training signal comes from correctness pairs, which establish a direct preference between correct and incorrect responses:

$$\begin{cases} c(y_w) = 1, \quad c(y_l) = 0 \\ r(y_w) \geq 1 \end{cases}, \tag{4}$$

where $c(\cdot)$ denotes the correctness score. The reasoning constraint $r(y_w) \geq 1$ filters out responses that arrive at the correct answer through flawed reasoning. As a complementary signal, when multiple candidates are all correct, we further construct quality pairs satisfying $c(y_w) = c(y_l) = 1$ and $r(y_w) - r(y_l) \geq 1$, encouraging the model to produce more rigorous arguments among correct solutions. For both types, the method score serves as a secondary ranking criterion, continuously reinforcing the strategy acquired in Stage 1. The C-DPO training objective is:

$$\mathcal{L}_{\text{C-DPO}} = -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}_2}\Bigg[\log\sigma\Bigg(\beta_c \log\frac{\pi_\theta(y_w \mid x)}{\pi_1(y_w \mid x)} - \beta_c \log\frac{\pi_\theta(y_l \mid x)}{\pi_1(y_l \mid x)}\Bigg)\Bigg], \tag{5}$$

where π_1 serves as both the initialization and the reference policy, β_c is the temperature parameter, and $\mathcal{D}_2$ denotes the correctness-oriented preference pairs augmented with a subset of Stage 1 pairs to prevent forgetting of the previously acquired reasoning strategy. The complete training pipeline is summarized in Algorithm 1.

Algorithm 1 Stage-wise Rubric Preference Training

Require: Problem set $\mathcal{D}$, base model π_0, ground-truth solutions $\mathcal{G}$, judge model $\mathcal{J}$
Ensure: Final model π_2
// Data Preparation
for each problem $d \in \mathcal{D}$ **do**
  Sample K responses from π_0 given d
  Generate rewrite responses from π_0 guided by $\mathcal{G}(d)$
  Score all responses with $\mathcal{J}$ on (m, r, c)
end for
// Stage 1: M-DPO (Method-Oriented Preference Learning)
for each problem $d \in \mathcal{D}$ **do**
  Construct pairs $\mathcal{D}_1(d)$ where $m(y_w) - m(y_l) \geq 1$, $r(y_w) \geq 1$, $r(y_l) \geq 1$; prioritize pairs with higher $r(y_w)$
end for
Train $\pi_1 \leftarrow$ M-DPO(π_0, $\bigcup_x \mathcal{D}_1(d)$)
// Stage 2: C-DPO (Correctness-Oriented Preference Refinement)
for each problem $d \in \mathcal{D}$ **do**
  Re-sample K responses from π_1 given d
  Score all responses with $\mathcal{J}$ on (m, r, c)
  Construct correctness pairs: $c(y_w)$=1 vs $c(y_l)$=0
  Construct quality pairs: both c=1, $r(y_w) - r(y_l) \geq 1$
end for
$\mathcal{D}_2 \leftarrow \bigcup_x \mathcal{D}_2(x) \cup$ Replay($\mathcal{D}_1$)
Train $\pi_2 \leftarrow$ C-DPO(π_1, $\mathcal{D}_2$)
return π_2

4 Experiments

4.1 Experimental Setup

Dataset. We select mathematical proof problems involving examples, counterexamples, or constructions from the training set of BrokenMath (Petrov et al., 2025), a theorem-proving benchmark of approximately 15K problems sourced from DeepTheorem and NuminaMath-1.5, as our training data. Candidate problems are identified via DeepSeek-R1 based on predefined criteria (Appendix B) and manually verified, yielding 1,275 problems. For each problem, we sample 16 candidate responses from the base model and generate 4 additional responses via RGSI. After preference pair construction, Stage 1 and Stage 2 each yield approximately 14,000 training pairs. The SFT stage is trained on ground-truth references as a cold start. We evaluate on CounterMath (Li et al., 2025b), which directly targets example-driven reasoning, using its four metrics: Macro-F1, Examples, Strict Align, and Loose Align (Appendix E).

Baselines. We compare against a broad range of models, including commercial models (e.g., Claude-4.5-Sonnet (Anthropic, 2025), Gemini 3 Pro (Google DeepMind, 2025)) and open-source models spanning 3B to 72B parameters, covering both general-purpose and math-specialized models. All baselines are listed in Table 2. We report progressive results (Base, +SFT, +Stage 1, +Stage 1+Stage 2) on all three configurations. This setup covers two model scales within the Qwen2.5-Math (1.5/7B) (Yang et al., 2024) and Llama-3.1-8B-Instruct (Grattafiori et al., 2024) as a cross-family configuration, allowing us to assess both scale and cross-family generalizability.

Implementation Details. We use Qwen2.5-Math-7B-Instruct as our primary base model and conduct all training with LLaMA-Factory (Zheng et al., 2024). Both the SFT and DPO stages employ LoRA (Hu et al., 2022). DeepSeek-V3.2 serves as the judge model for rubric scoring. During inference, we use vLLM (Kwon et al., 2023) for generation with greedy decoding for all evaluations. All experiments are conducted on 4 NVIDIA L20 GPUs. We additionally apply the same pipeline to Qwen2.5-Math-1.5B-Instruct with identical hyperparameters. To further verify cross-family generalizability, we apply the same pipeline and hyperparameters to Llama-3.1-8B-Instruct. Full hyperparameter settings are provided in Appendix E.

Table 2: Main evaluation results on CounterMath. Results are grouped into commercial models, open-source models, and our method with progressive training stages. Bold indicates the best result within each group.

Models	Judgement F1 (macro)	Rationale Reasoning Examples (%)	Strict (%)	Loose (%)
Commercial models				
KIMI-K2.5 (Bai et al., 2026)	51.79	71.63	34.78	45.80
GLM-4.6 (Zeng et al., 2025)	58.39	73.19	41.45	45.06
Claude-4.5-Sonnet (Anthropic, 2025)	74.24	72.94	39.30	45.64
DeepSeek-R1 (Guo et al., 2025)	73.50	80.59	42.52	46.95
Gemini-3-Pro (Google DeepMind, 2025)	**80.43**	**81.17**	**45.56**	**50.82**
Open-source models				
Llama-3.2-3B-Instruct (Grattafiori et al., 2024)	35.78	66.53	4.28	4.76
Phi-4-mini-3.8B-instruct (Abouelenin et al., 2025)	29.67	49.51	7.81	9.04
Qwen3-4B (Yang et al., 2025)	33.95	58.14	14.47	17.93
DeepSeek-Math-7B-rl (Shao et al., 2024)	33.91	65.46	10.19	10.94
Eurus-2-7B-PRIME (Cui et al., 2025)	36.75	62.83	14.63	17.35
Qwen3-32B (Yang et al., 2025)	**45.93**	**76.97**	**25.41**	**30.67**
Qwen2.5-Math-72B-Instruct (Yang et al., 2024)	41.89	75.99	21.46	24.67
INSPIRE (Ours)				
Llama-3.1-8B-Instruct (Grattafiori et al., 2024)	41.28	76.73	6.57	7.56
+ SFT	42.98 (+1.70)	80.10 (+3.37)	7.40 (+0.83)	8.31 (+0.75)
+ Stage 1	48.15 (+6.87)	89.88 (+13.15)	8.22 (+1.65)	9.78 (+2.22)
+ Stage 1 + Stage 2 (Full)	**50.27** (+8.99)	**93.09** (+16.36)	**9.45** (+2.88)	**11.80** (+4.24)
Qwen2.5-Math-1.5B-Instruct (Yang et al., 2024)	35.71	69.24	9.04	10.60
+ SFT	36.36 (+0.65)	69.82 (+0.58)	9.45 (+0.41)	10.93 (+0.33)
+ Stage 1	37.26 (+1.55)	74.84 (+5.60)	10.85 (+1.81)	11.84 (+1.24)
+ Stage 1 + Stage 2 (Full)	**39.02** (+3.31)	**76.32** (+7.08)	**11.68** (+2.64)	**13.49** (+2.89)
Qwen2.5-Math-7B-Instruct (Yang et al., 2024)	38.67	75.25	14.31	16.61
+ SFT	39.06 (+0.39)	77.63 (+2.38)	14.39 (+0.08)	17.02 (+0.41)
+ Stage 1	43.05 (+4.38)	81.33 (+6.08)	16.53 (+2.22)	19.08 (+2.47)
+ Stage 1 + Stage 2 (Full)	**45.91** (+7.24)	**84.79** (+9.54)	**17.30** (+2.99)	**20.07** (+3.46)

4.2 Main Results

Table 2 presents the main results on CounterMath. Across all three model configurations, each training stage yields consistent improvements. On the 7B scale, Stage 1 increases Examples from 77.63% to 81.33% and F1 from 39.06 to 43.05, confirming that method-oriented preference learning encourages example-based reasoning adoption. Stage 2 further raises F1 to 45.91 and Examples to 84.79%, demonstrating effective correctness refinement atop the acquired strategy. The alignment metrics improve steadily across stages, though absolute scores remain moderate since models may construct valid yet distinct examples that differ from references, as also noted in the CounterMath analysis, we therefore focus on relative improvement across stages. The same progressive pattern holds on both the 1.5B scale (F1: +2.66, Ex.: +7.08) and Llama-3.1-8B-Instruct (F1: +7.29, Ex.: +12.99), confirming the consistency of our approach across model scales and families. SFT provides marginal gains, primarily serving as format-level cold start.

Compared with open-source models, our 7B model **surpasses Qwen2.5-Math-72B-Instruct** (F1: 41.89) and approaches Qwen3-32B (F1: 45.93). Notably, smaller models generally exhibit lower Examples rates with correspondingly lower F1, suggesting that example usage is a key bottleneck for this task. More broadly, F1 and Examples are strongly correlated across models, confirming that actively employing examples is a primary driver of judgment accuracy on CounterMath. Our models also achieve **higher Examples rates than all other evaluated models**: 84.79% (Qwen-7B) and 93.09% (Llama-8B), both exceeding DeepSeek-R1 (80.59%). Furthermore, the 1.5B model after full training (F1: 39.02) **reaches comparable performance to the 7B base model** (F1: 38.67), demonstrating that our method can effectively compensate for the capacity gap through targeted training. The remaining gap to commercial models is largely attributable to base model capacity. Ablation studies and analyses in Sections 4.4 and 4.5 further validate each design choice.

Table 3: Out-of-distribution evaluation results on general mathematical reasoning benchmarks. Values in parentheses indicate the difference relative to the base model.

Models	GSM8K	MATH500	AIME 2024	GAOKAO-mQA	MMLU-cMath
Llama-3.1-8B-Instruct	82.94	47.80	6.67	32.48	42.00
+ Stage 1	83.40 (+0.46)	49.00 (+1.20)	6.67 (+0.00)	34.76 (+2.28)	43.00 (+1.00)
+ Stage 1 + Stage 2	**83.47** (+0.53)	**49.60** (+1.80)	6.67 (+0.00)	**35.04** (+2.56)	**45.00** (+3.00)
Qwen2.5-Math-1.5B-Instruct	85.22	73.80	10.00	73.22	70.00
+ Stage 1	85.22 (+0.00)	**75.20** (+1.40)	10.00 (+0.00)	73.79 (+0.57)	70.00 (+0.00)
+ Stage 1 + Stage 2	**85.75** (+0.53)	75.00 (+1.20)	10.00 (+0.00)	**74.83** (+1.61)	**73.00** (+3.00)
Qwen2.5-Math-7B-Instruct	94.54	84.20	13.33	78.06	72.00
+ Stage 1	95.45 (+0.91)	84.80 (+0.60)	16.67 (+3.34)	79.77 (+1.71)	73.00 (+1.00)
+ Stage 1 + Stage 2	**95.68** (+1.14)	**84.80** (+0.60)	**20.00** (+6.67)	**80.06** (+2.00)	**75.00** (+3.00)

Table 4: Ablation on training strategy. The first group compares alternative training strategies; the second group ablates individual stages of our pipeline.

Strategy	F1	Ex.	Str.	Loo.
Base (+ SFT)	39.06	77.63	14.39	17.02
DPO	39.38	73.19	15.21	17.11
Chosen SFT	41.75	79.52	15.04	18.09
Mixed DPO	42.72	81.17	15.21	17.92
C-DPO	40.27	77.88	15.13	17.43
M-DPO	43.05	81.33	16.53	19.08
Ours(Full)	**45.91**	**84.79**	**17.30**	**20.07**

Table 5: Ablation on response augmentation strategy in Stage 1. *Self-sampling* generates candidates without guidance; *Teacher rewriting* uses DeepSeek-R1 to rewrite references.

Strategy	F1	Ex.	Str.	Loo.
Base (+ SFT)	39.06	77.63	14.39	17.02
Self-sampling	40.40	78.55	15.54	17.68
Teacher rewriting	39.14	74.01	15.21	17.51
RGSI (Ours)	**43.05**	**81.33**	**16.53**	**19.08**

4.3 Out-of-Distribution Evaluation

To examine the broader impact of our training on general mathematical reasoning, we evaluate on several out-of-distribution benchmarks including GSM8K, MATH500, AIME 2024, GAOKAO-mathQA, and MMLU-collegeMath. As shown in Table 3, the final model consistently maintains or improves performance across all benchmarks on all three model configurations. The unchanged AIME 2024 scores for the two smaller configurations are expected given its 30-problem granularity and competition-level difficulty; notably, Qwen2.5-Math-7B achieves +6.67 on this benchmark. These results suggest that our specialized training may also benefit general mathematical reasoning through improved analytical strategies.

4.4 Ablation Studies

All ablation experiments are conducted on Qwen2.5-Math-7B-Instruct.

Effect of Training Strategy. We compare against several alternative strategies in Table 4. DPO, the standard preference optimization setup using answer correctness as the sole signal, yields minimal F1 improvement and actually reduces example usage (73.19% vs. 77.63%), indicating that optimizing solely for correctness can suppress example-based reasoning. Chosen SFT, trained on all chosen responses from both stages, achieves moderate gains (F1: 41.75) but falls short of our full pipeline (45.91), confirming that contrastive preference signals outperform imitation alone. Mixed DPO, trained on the shuffled union of Stage 1 and Stage 2 preference pairs in a single pass, underperforms even Stage 1 alone (42.72 vs. 43.05) despite using more data, suggesting that method and correctness objectives introduce conflicting gradients when mixed. Among stage-wise ablations, M-DPO alone (43.05) substantially outperforms C-DPO alone (40.27), confirming that method-oriented training is the critical first step. The full sequential pipeline consistently yields the best results across all evaluated metrics.

Effect of Reference-Guided Student Internalization. We compare three response augmentation strategies for constructing preference pairs in Stage 1: (1) self-sampling, where all candidates are generated by the base model without any guidance; (2) teacher rewriting, where DeepSeek-R1 directly rewrites the reference solutions; and (3) our proposed RGSI, where the student model regenerates responses after comprehending the reference.

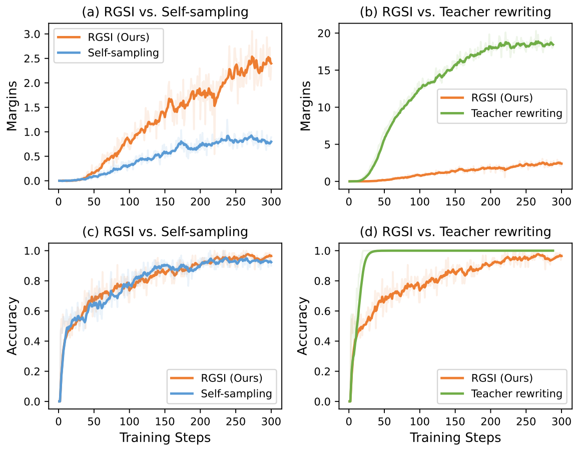

Figure 3: DPO training dynamics under different augmentation strategies. RGSI maintains moderate margins and steady gradual accuracy growth, while self-sampling yields insufficient margins and teacher rewriting leads to trivially separable pairs.

As shown in Table 5, self-sampling yields only marginal improvement, as candidates lack sufficient quality variation to form effective preference pairs. Teacher rewriting, despite producing high-quality responses, nonetheless fails to bring improvement due to distribution shift between the teacher and student generation styles.

Figure 3 further illustrates this from the training dynamics perspective. Self-sampling results in very small margins (a), indicating limited quality differences between candidates. Teacher rewriting produces excessively large margins (b) and near-instant accuracy saturation (d), suggesting the model distinguishes pairs by surface-level style rather than reasoning quality. RGSI maintains moderate margins (a) with gradual accuracy growth (c), providing meaningful learning signals throughout.

4.5 Analysis

Training vs. Hint Prompting. As shown in Table 6, hint prompting increases example usage but yields only marginal improvement in judgment accuracy (F1: +0.46), suggesting that elicited examples are not well-targeted enough to support correct reasoning. In contrast, our training approach achieves substantially larger gains across all metrics (F1: +7.24), indicating the model learns to construct appropriate examples that effectively contribute to sound judgments.

Method Acquisition Analysis. To quantify the behavioral shift induced by Stage 1, we compare the method score distribution of sampled responses from π_0 and π_1 (Appendix F). After Stage 1, substantive example usage (m = 2) increases from 30.74% to 39.30%, while purely abstract reasoning (m = 0) decreases from 56.58% to 48.47%, confirming that method-oriented training shifts the model toward example-based reasoning.

Table 6: Comparison between hint prompting and our stage-wise training approach.

Setting	F1	Ex.	Str.	Loo.
Base	38.67	75.25	14.31	16.61
Base + Hint prompt	39.13	77.71	15.95	18.25
Ours (Full)	**45.91**	**84.79**	**17.30**	**20.07**

Table 7: Cross-judge evaluation on Qwen2.5-Math-7B-Instruct. F1 is judge-independent; Ex., Str., and Loo. are evaluated by GPT-4o.

Stage	F1	Ex.	Str.	Loo.
Base	38.67	78.21	32.81	37.83
+ SFT	39.06	79.11	33.88	38.89
+ Stage 1	43.05	86.10	37.34	42.19
+ Stage 1 + Stage 2	**45.91**	**87.50**	**39.56**	**43.67**

Cross-Judge Validation. We adopt DeepSeek-V3.2 as our judge model, favoring an open-source model for reproducibility. To rule out judge bias, we re-evaluate all stages using GPT-4o. As shown in Table 7, although absolute scores shift due to judge calibration, the progressive improvement across stages holds consistently, confirming our results are not an artifact of a specific judge.

Case Study. A case study in Appendix F illustrates the progressive shift from abstract reasoning to targeted counterexample construction.

5 Conclusion

In this paper, we propose INSPIRE, an Internalize-Then-Improve approach that enhances example-driven mathematical reasoning through Reference-Guided Student Internalization and stage-wise rubric preference training. Experiments across multiple model scales and families demonstrate consistent improvements, with maintained or improved performance on out-of-distribution benchmarks. Our results confirm that decomposing capability acquisition into strategy adoption and correctness refinement is more effective than joint optimization. Future work may explore scaling to larger datasets and models, as well as extending the approach to other underexplored reasoning strategies.

Limitations

Our training data consists of 1,275 problems from BrokenMath, covering a limited set of mathematical domains. Whether the framework generalizes to larger-scale data and broader areas such as number theory and combinatorics remains to be explored. We validate on models up to 7B parameters; the effectiveness on larger scales (e.g., 32B or 72B) has not been examined. Both data construction and evaluation rely on DeepSeek-V3.2 as the judge model, which may introduce biases; we mitigate this through cross-validation with GPT-4o, confirming consistent trends. All reported results are from single runs with fixed random seeds, though the consistent improvements across three model configurations suggest the findings are robust.

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A Evaluation Rubric Prompts

We provide the prompt used for scoring candidate responses along the Method and Reasoning dimensions defined in Table 1. Each response is scored independently by the judge model using the following prompt. The Correctness dimension is determined by exact matching between the model's final answer and the ground-truth label.

Scoring Prompt

[Task]
Evaluate the given mathematical proof. Score independently on two dimensions:
1. **Method**: Whether the proof uses concrete examples, counterexamples, or constructions, and whether these instances play a substantive role in the reasoning
2. **Reasoning**: Whether the logical structure of the proof is complete and clear

Strictly follow the steps below to analyze before scoring. Do not skip any analysis step.
[Method Dimension: Use of Examples / Counterexamples / Constructions]
Evaluation Steps
Step 1: Identify Concrete Mathematical Objects
Find all concrete mathematical objects in the proof, including:
- Specific values (e.g., n=1, x=187)
- Specific functions (e.g., $f(x)=|x|$, $f(x)=x^2$)
- Specific sets, groups, or spaces (e.g., $Z_2 \times Z_2$, A={{1,2,3}})
- Specific matrices, geometric figures, sequences, etc.
Focus on whether the proof **constructs or provides concrete objects** to support the argument, rather than merely performing algebraic simplification or mentioning abstract objects (e.g., "for any x", "let G be a group", etc.)
If no concrete objects are found, let method = 0, skip directly to the Reasoning dimension.

Step 2: Assess Contribution
For each concrete object found, determine whether it plays a direct and critical role in the reasoning.
Specifically, determine whether the object is used to:
(a) Prove existence — providing a concrete witness or enumeration verification
(b) Disprove a universal claim — providing a counterexample to refute a proposition
(c) Reveal key properties — using a concrete instance to demonstrate critical properties or behaviors of mathematical objects
(d) Advance reasoning steps — using computation on a concrete instance to drive a key step of the proof
A critical role means: the concrete object provides **specific insight that purely abstract description cannot easily convey**, rather than merely serving as a supplementary illustration of an already-complete abstract argument.

The following do **not** count as direct contributions:
- Mentioning a concrete object only for background introduction or incidental illustration
- Adding an example after an abstract proof is already complete, just to explain the conclusion
- The object is dispensable in the proof, removing it does not affect the completeness of the argument

Step 3: Scoring
- **2**: Concrete objects exist and play a direct critical role in at least one of (a)(b)(c)(d) above
- **1**: Concrete objects exist but are used only for incidental illustration or auxiliary explanation, not playing a direct role in the key reasoning
- **0**: No concrete objects; purely abstract reasoning

[Reasoning Dimension: Reasoning Quality]
Evaluate the logical coherence and completeness of the reasoning:
- **3**: Every step is well-justified; the argument is complete and clearly written
- **2**: Overall logic is clear, but some details have minor issues
- **1**: Has a proof framework, but key steps are missing or contain gaps
- **0**: Logic is chaotic, contains obvious contradictions, or text is severely corrupted

[Output Format]
Output in the following format:
Method Analysis:
1. Concrete objects: [List all concrete mathematical objects found, or "None"]
2. Contribution assessment: [For each object, state which category (a)(b)(c)(d) it belongs to,
and whether it plays a direct critical role in the proof]
3. Method score rationale: [One sentence summarizing why this score is given]
Reasoning Analysis:
[Briefly evaluate the logical structure and completeness of the reasoning, noting main strengths or issues]
Scores:
{{"method": <0-2>, "reasoning": <0-3>}}

[Input]
{statement}
{rationale}

B Problem Filtering Criteria

We use DeepSeek-R1 to identify candidate problems from BrokenMath that involve examples, counterexamples, or constructions. The filtering prompt is shown below.

Filtering Prompt

```
# Instruction
You are an expert in mathematics, particularly skilled at handling and analyzing mathematical proofs. Please assess whether the following mathematical proof or explanation uses examples or counterexamples that directly contribute to understanding concepts or validating arguments.

# Evaluation Criteria
Evaluate based on these criteria:

## Criterion 1: Concrete Instance Construction
Does the proof/explanation construct or provide specific, concrete examples or counterexamples?
- This includes: specific numbers, functions, matrices, sets, geometric figures, or other mathematical objects.
- Focus on whether the content **constructs or provides** concrete instances, not just mentions abstract cases.
- Example: "Let n=1", "Consider f(x)=|x|", "Take matrix A=[[1,0],[0,1]]", etc

## Criterion 2: Direct Contribution to Understanding or Validation
Does the constructed example or counterexample directly help in understanding the concept or validating the argument?
- The example should play a meaningful role in: (a) Clarifying mathematical concepts (b) Verifying or refuting claims
- "Direct contribution" means: the example provides concrete insight that abstract description alone cannot easily convey.
- Avoid: mere casual mentions or vague applications without specific construction.

## Criterion 3: Essential Role in Reasoning
Does the example serve one of these essential purposes?
- Demonstrates existence (for existence claims)
- Disproves universality (counterexample for universal claims)
- Illustrates key properties or behaviors of mathematical objects
- Validates intermediate steps through concrete computation

**Note**: Criterion 3 represents concrete manifestations of "direct contribution" in Criterion 2. When evaluating Criterion 2, refer to the four purposes listed in Criterion 3 to determine if the example truly plays an important role.

# Decision Rule
Mark as **True** if:
- Both Criterion 1 and Criterion 2 are satisfied
Mark as **False** if:
- The proof relies purely on abstract logical reasoning, algebraic manipulation, or theorem citations without concrete examples
- The example, though concrete, does not substantially help understanding or validation (fails all items in Criterion 3)

# Output Format
Please output strictly in the following JSON format without any additional content:
{{
  "reason": "State whether an example or counterexample was provided and whether it meets the evaluation criteria. If not, provide the reason.",
  "is_suitable": true or false
}}

# Input
**Statement**:
{statement}

**Rationale**:
{rationale}
```

C Reference-Guided Student Internalization

In RGSI, the base model is provided with the reference solution and asked to regenerate a complete response in its own expression style. The rewriting prompt is shown below.

Rewriting Prompt

Task
Solve the following mathematics problem independently.

Problem
{statement}

Hint
{hint}

Instructions
- Start by analyzing the problem conditions and structure step by step.
- Use the hint as inspiration for your approach, but write the full solution in your own words.
- Show complete step by step reasoning without directly stating or relying on the hint's conclusion in your answer.
- Provide your solution directly with no preamble, and place your final answer in \boxed{{}}.

Output

D Inference Prompt Templates

We use two inference prompts in our evaluation. The standard prompt is used for all main experiments, and the hint prompt explicitly requests example-based reasoning, as compared in Section 4.5.

Standard Prompt

{statement}
Please reason step by step about whether the above statement is True or False, and put your final answer within \\boxed{}.

Hint Prompt

{statement}
Please reason by giving examples about whether the above statement is True or False, and put your final answer within \\boxed{}.

E Training and Details

Evaluation Metrics. We adopt the evaluation metrics from CounterMath (Li et al., 2025b): **Macro-F1** measures the accuracy of the model's true/false judgments on mathematical statements, using macro averaging to account for label imbalance; **Examples** measures the proportion of problems where the model employs examples during reasoning; **Strict Align** measures the proportion of model-provided examples that are fully consistent with the reference; **Loose Align** measures the proportion of instances where at least one model-provided example is consistent with the reference. We use DeepSeek-V3.2 as the judge model for computing these metrics, with the same evaluation prompts as CounterMath.

Hyperparameters. For SFT, we set the learning rate to 1×10^{-5} and train for 3 epochs. For DPO, we set the learning rate to 5×10^{-6}, $\beta = 0.2$, and train for 3 epochs. The same hyperparameters are used for both model scales.

F Additional Analysis

Method Score Distribution. For each of the 1,275 problems, we sample 16 responses from both π_0 and π_1 under the same settings, yielding 20,400 self-sampled responses per model without RGSI responses. After DeepSeek-V3.2 scoring and parsing, 20,245 responses from π_0 and 20,171 from π_1 remain valid (155 and 229 excluded, respectively). Table 8 reports the resulting distributions.

Table 8: Method-score distribution of self-sampled responses before and after Stage 1.

Model	Valid responses	$m = 0$	$m = 1$	$m = 2$
π_0 (before Stage 1)	20,245	11,455 (56.58%)	2,567 (12.68%)	6,223 (30.74%)
π_1 (after Stage 1)	20,171	9,777 (48.47%)	2,466 (12.23%)	7,928 (39.30%)

After Stage 1, substantive example usage ($m = 2$) increases from 30.74% to 39.30%, while purely abstract reasoning ($m = 0$) decreases from 56.58% to 48.47%.

Case Study. Figure 4 presents a representative example from CounterMath. The base model π_0 applies the continuity-from-above theorem without examining its precondition ($m(E_1) < \infty$), leading to an incorrect judgment through purely abstract reasoning. After Stage 1 training, π_1 learns to actively construct examples, but the chosen example satisfies the theorem's precondition and merely confirms rather than challenges it. After Stage 2 training, π_2 identifies the critical precondition and deliberately constructs a counterexample that violates it ($E_n = [n, \infty)$), arriving at the correct answer. This progression demonstrates that Stage 1 equips the model with the *strategy* of using examples, while Stage 2 refines the *quality* by teaching it to construct targeted examples that test theorem boundaries.

Case of progressive improvement across training stages on concept-level reasoning

Statement: "Existing a decreasing sequence of measurable sets ({E_n}) such that (m(\lim_{n \to \infty} E_n) \neq \lim_{n \to \infty} m E_n)."
Answer: True

Base Model π_0 **Prediction:** False

Key Excerpt: "...By the properties of measure, for a decreasing sequence of measurable sets, we have (m(\bigcap_{n=1}^{\infty} E_n) = \lim_{n \to \infty} m(E_n)). This means that the measure of the limit of the sequence is equal to the limit of the measures ... Therefore, for any decreasing sequence of measurable sets ({E_n}), we **always** have (m(\lim_{n \to \infty} E_n) = \lim_{n \to \infty} m(E_n)). ...The answer is **False.**"
Key Excerpt: The model applies the continuity-from-above theorem **without examining its precondition** ((m(E_1) < \infty)) and provides **no concrete example**, relying entirely on abstract theorem application.

Stage 1 Model π_1 **Prediction:** False

Key Excerpt: "...The Lebesgue measure is continuous from above, meaning (m(\bigcap_{n=1}^{\infty} E_n) = \lim_{n \to \infty} m(E_n)). ...**To confirm this, let's consider a specific example.** Suppose (E_n = [0, 1 - \frac{1}{n}]). The intersection of all these sets is (\bigcap_{n=1}^{\infty} E_n = [0, 1)). The measure of ([0,1)) is 1. The limit of the measures is (\lim_{n \to \infty}(1 - \frac{1}{n}) = 1). Thus (m(\bigcap_{n=1}^{\infty} E_n) = 1 = \lim_{n \to \infty} m(E_n)). ...The answer is **False**."
Key Excerpt: Compared to the base model, π_1 **learns to actively construct examples** (using (E_n = [0, 1-\frac{1}{n}])). However, the selected example satisfies (m(E_1) < \infty), **which merely confirms the theorem rather than testing its boundary**. The model uses examples to verify rather than to challenge.

Stage 2 Model π_2 **Prediction:** True

Key Excerpt: "...The Lebesgue measure is continuous from above, meaning (m(\bigcap_{n=1}^{\infty} E_n) = \lim_{n \to \infty} m(E_n)), **provided that (m(E_1) < \infty). However, if (m(E_1) = \infty), the continuity from above may not hold. Let's construct an example.** Consider (E_n = [n, \infty)). The intersection is (\bigcap_{n=1}^{\infty}[n, \infty) = \emptyset), so (m(\bigcap_{n=1}^{\infty} E_n) = 0). On the other hand, (m(E_n) = \infty) for all (n), so (\lim_{n \to \infty} m(E_n) = \infty). Since (0 \neq \infty), ...The answer is **True**."
Analysis: Compared to π_1, π_2 **identifies the critical precondition** (m(E_1) < \infty) that both previous stages overlooked, and **deliberately constructs an example that violates this condition** ((E_n = [n, \infty))). This demonstrates a shift from using examples to confirm theorems to using examples to expose their limitations.

Summary: Base model π_0 relies on pure theorem application without examples; Stage 1 training enables active example construction but examples only confirm existing theorems; Stage 2 training further enables the model to identify theorem preconditions and construct targeted counterexamples, leading to correct reasoning.

Figure 4: Case study comparing reasoning across training stages on a measure theory problem from CounterMath.

G Detailed Rubric Criteria

Table 9 provides the full scoring criteria for each dimension and score level defined in our evaluation rubric (Table 1).

Dimension	Score	Criteria
Method (m)	2	Concrete mathematical objects (specific values, functions, sets, matrices, etc.) are constructed and play a **direct critical role** in the reasoning: proving existence, disproving a universal claim, revealing key properties, or advancing a key reasoning step, et al.
	1	Concrete objects are present but serve only as incidental illustration or auxiliary explanation, without playing a direct role in the core argument.
	0	No concrete objects; the response relies entirely on abstract reasoning or theorem application.
Reasoning (r)	3	Every step is well-justified; the argument is complete and clearly written.
	2	Overall logic is clear, but some details have minor issues.
	1	Has a proof framework, but key steps are missing or contain logical gaps.
	0	Logic is chaotic, contains obvious contradictions, or text is severely corrupted.
Correctness (c)	1	The final answer matches the ground-truth label.
	0	The final answer does not match the ground-truth label.

Table 9: Detailed scoring criteria for each dimension and score level of the evaluation rubric.

H Additional Experiments and Reproducibility Details

H.1 Evaluation on a Recent Backbone

To examine whether INSPIRE remains effective on a more recent backbone, we apply the full two-stage pipeline to Qwen3-4B-Instruct-2507(Yang et al., 2025) following the same training and evaluation pipeline as in our main experiments.

Table 10: Evaluation of INSPIRE on the recent Qwen3-4B-Instruct-2507 backbone. Bold indicates the best result.

Stage	F1	Ex.	Str.	Loo.
Base	33.65	65.21	18.83	25.65
+ SFT	33.71	65.79	19.32	26.39
+ Stage 1	35.35	73.03	20.23	27.06
+ Stage 1 + Stage 2	**36.36**	**74.34**	**21.96**	**29.03**

H.2 Correctness-Only DPO Sensitivity

We train the correctness-only DPO baseline from the same SFT checkpoint and candidate pool as our main pipeline, obtaining 10,539 preference pairs based solely on final-answer correctness. All variants use LoRA with a learning rate of 5×10^{-6} for three epochs, while β is varied with all other settings fixed.

Table 11: Sensitivity of the correctness-only DPO baseline to β on Qwen2.5-Math-7B-Instruct. Bold indicates the best result.

Setting	F1	Ex.	Str.	Loo.
Base (+ SFT)	39.06	77.63	14.39	17.02
DPO ($\beta = 0.1$)	38.90	73.27	15.87	17.51
DPO ($\beta = 0.2$)	39.38	73.19	15.21	17.11
DPO ($\beta = 0.5$)	38.94	74.42	13.56	15.71
Ours (Full)	**45.91**	**84.79**	**17.30**	**20.07**

Correctness-only DPO remains close to the SFT baseline across all values of β and consistently reduces example usage, suggesting that its limited effectiveness is not specific to a particular regularization setting.